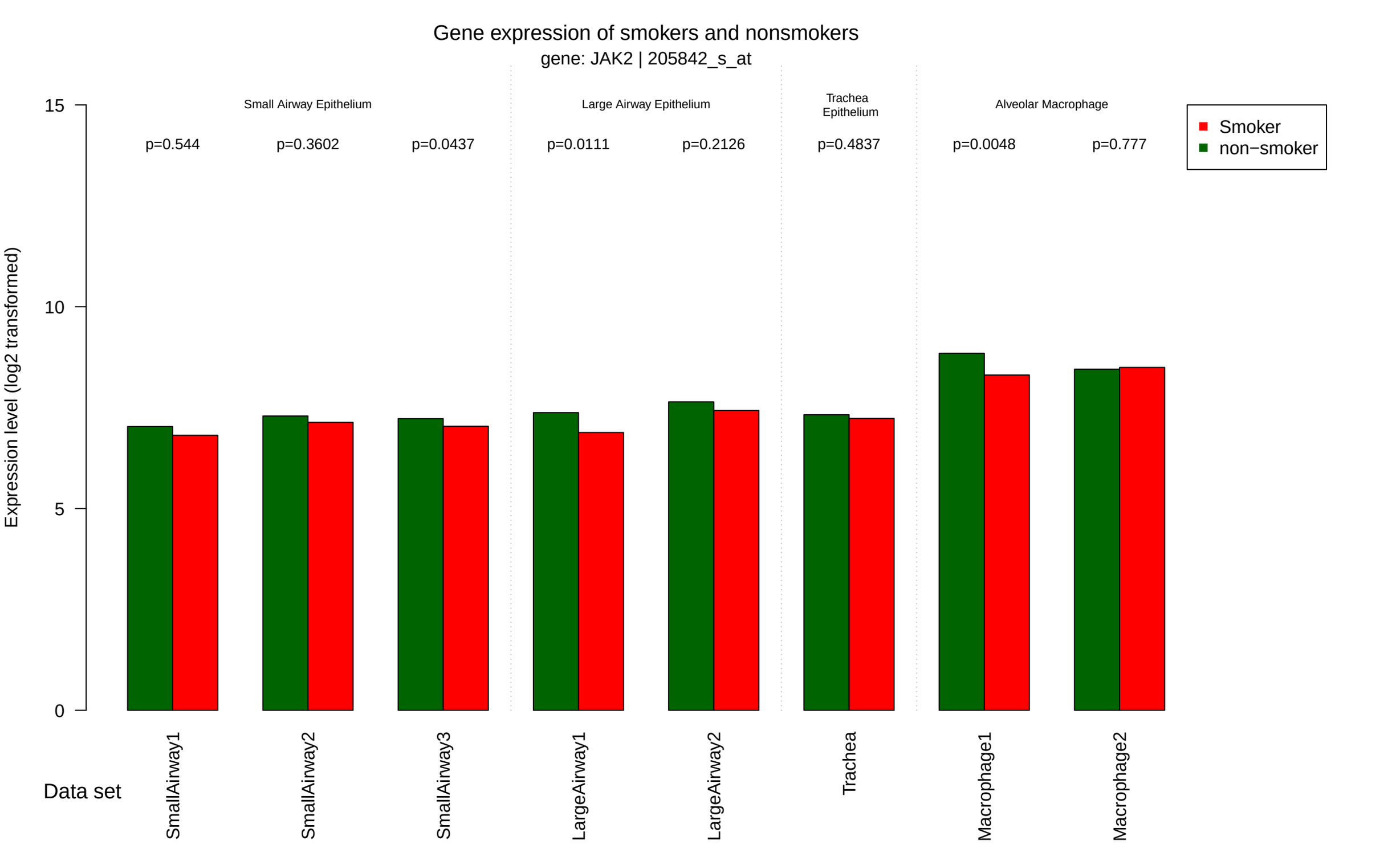
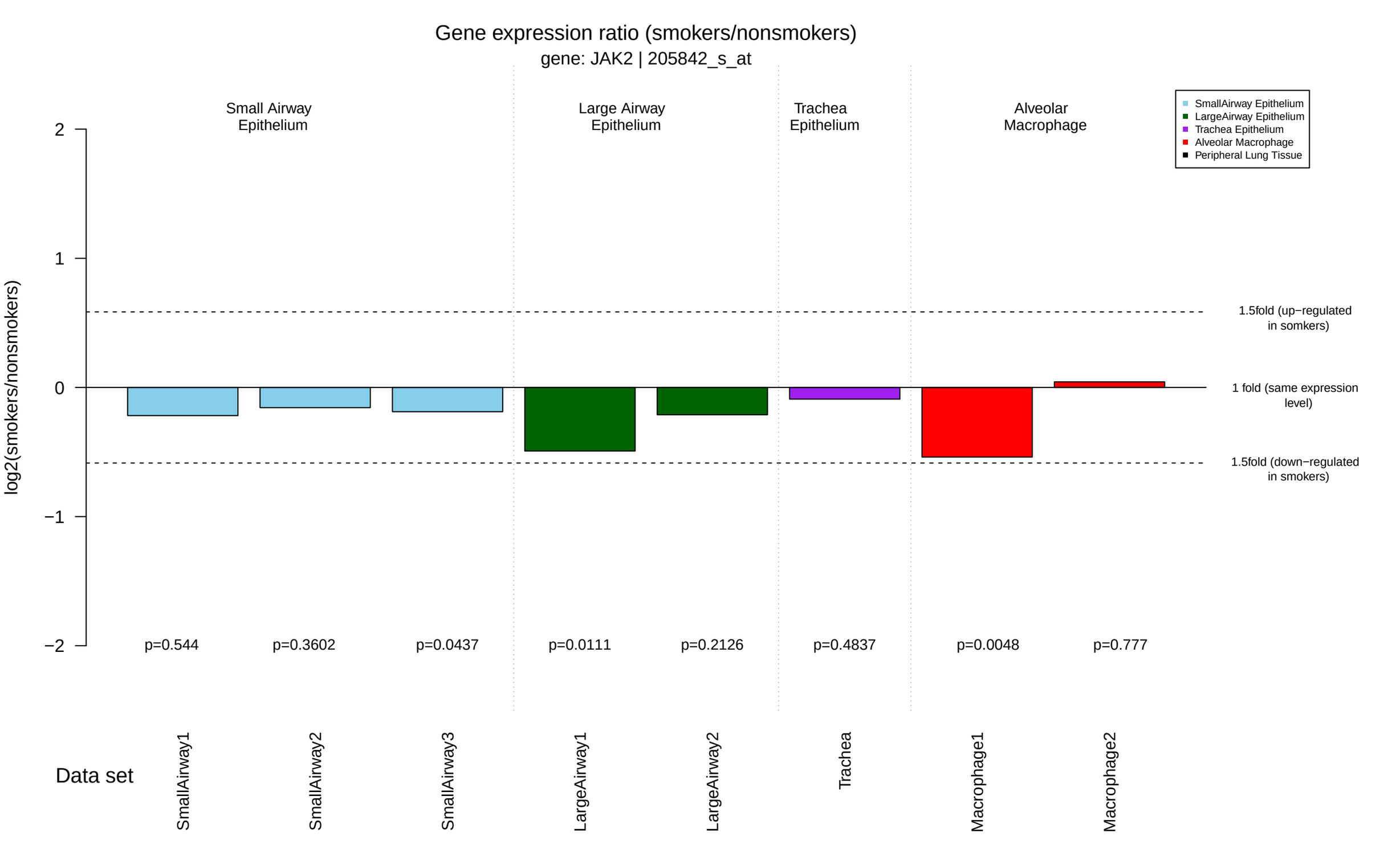
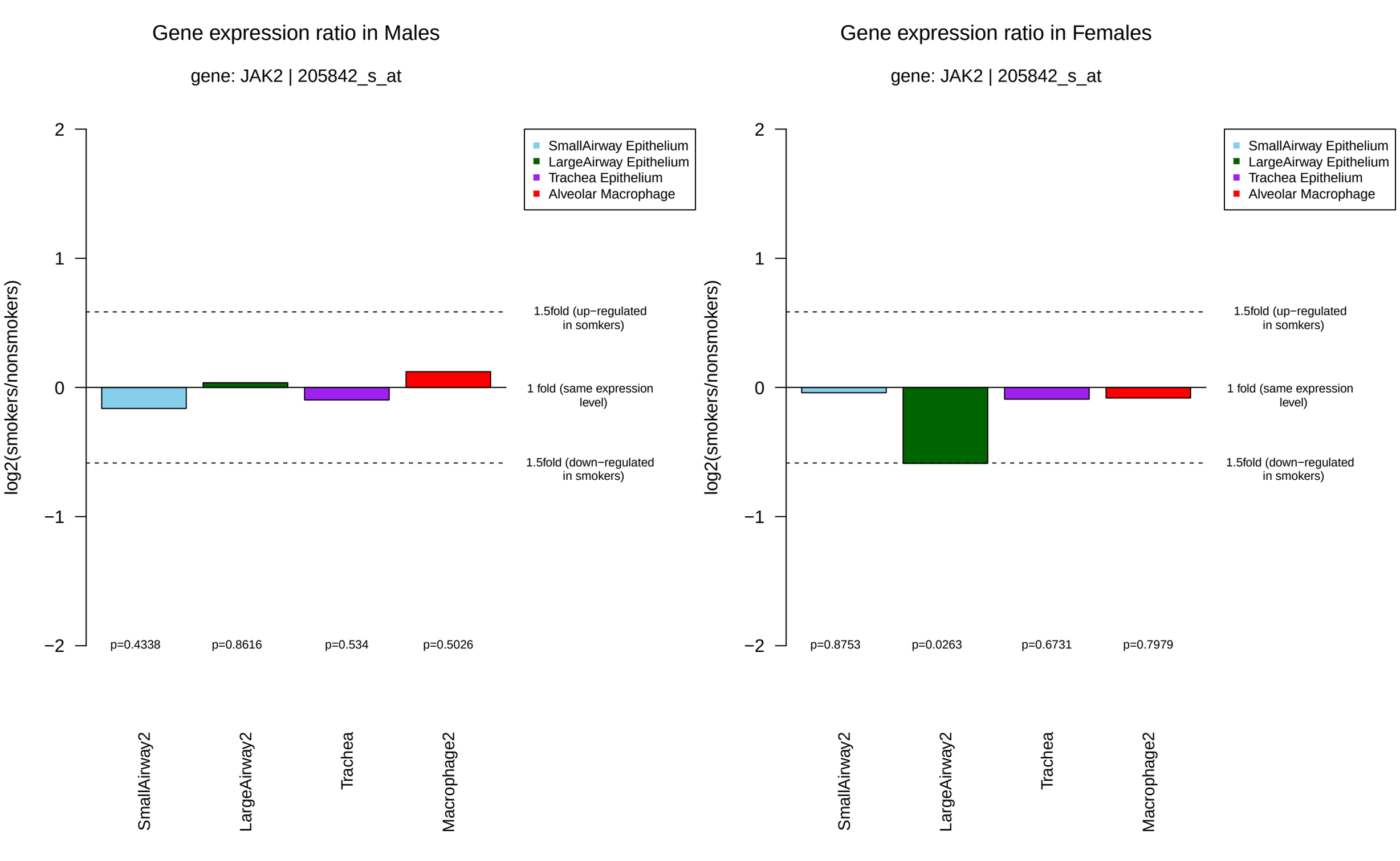
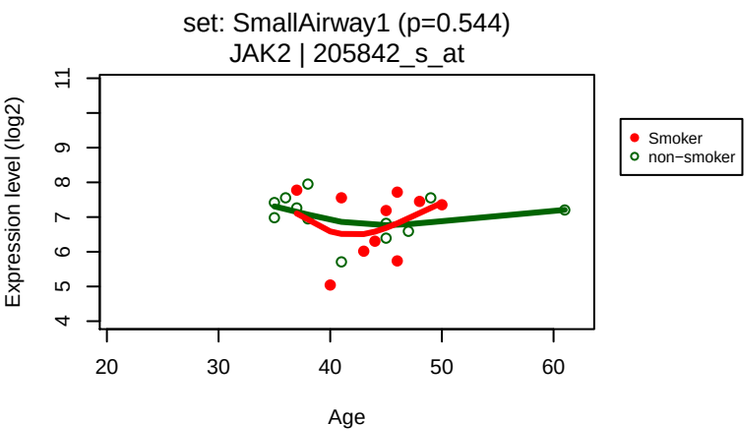
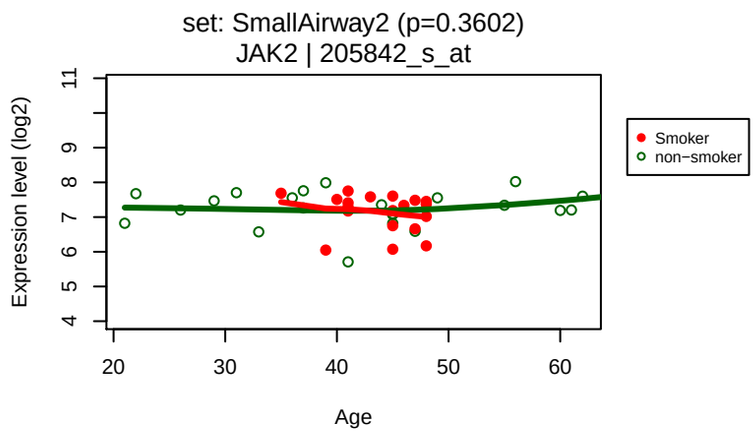
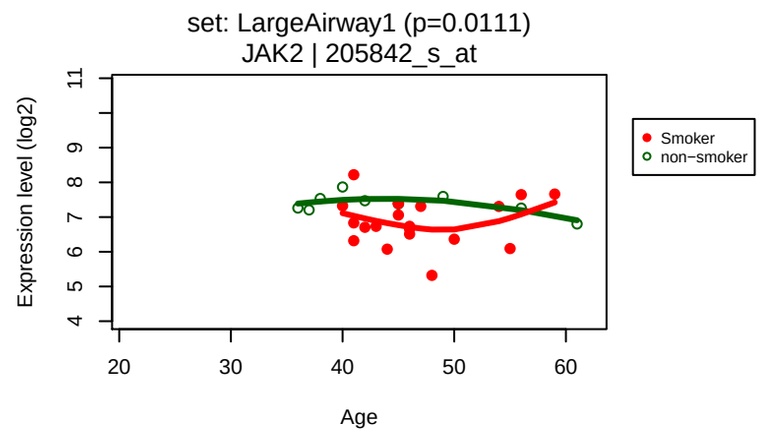

Gene expression of smokers and nonsmokers
gene: JAK2 | 205842_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.544
p=0.3602
p=0.0437
p=0.0111
p=0.2126
p=0.4837
p=0.0048
p=0.777
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

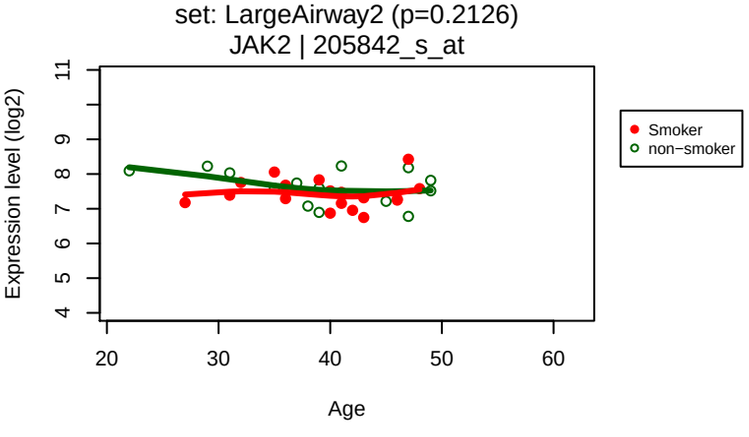

Gene expression ratio (smokers/nonsmokers)
gene: JAK2 | 205842_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.544
p=0.3602
p=0.0437
p=0.0111
p=0.2126
p=0.4837
p=0.0048
p=0.777
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

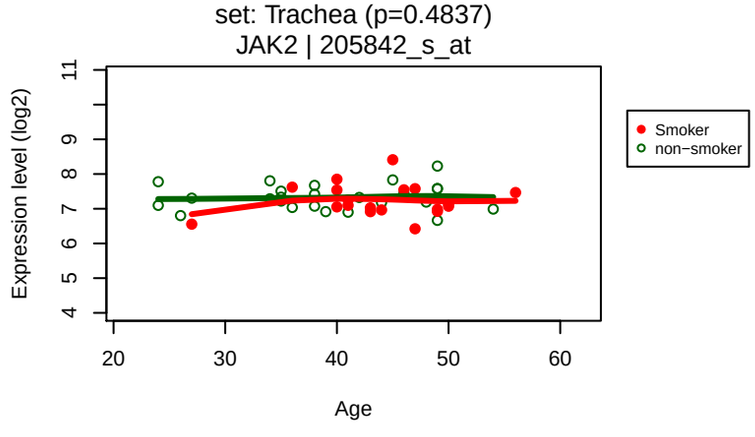

Gene expression ratio in Males
gene: JAK2 | 205842_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4338
p=0.8616
p=0.534
p=0.5026
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: JAK2 | 205842_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8753
p=0.0263
p=0.6731
p=0.7979
SmallAirway2
LargeAirway2
Trachea
Macrophage2

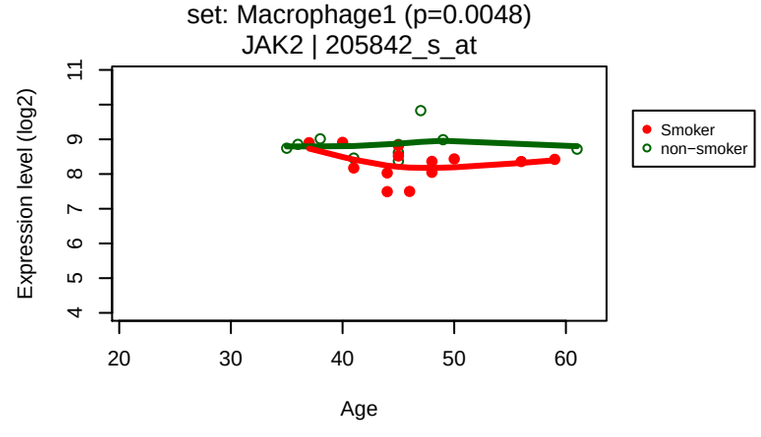

set: SmallAirway1 (p=0.544)
JAK2 | 205842_s_at
Expression level (log2)
Age
Smoker
non−smoker

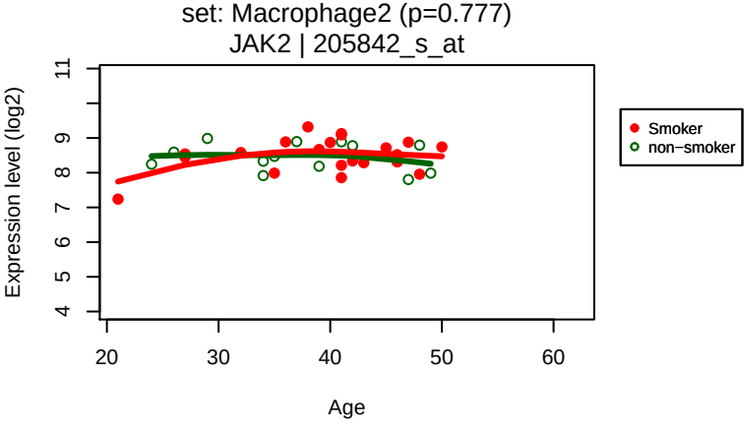

set: SmallAirway2 (p=0.3602)
JAK2 | 205842_s_at
Expression level (log2)
Age
Smoker
non−smoker

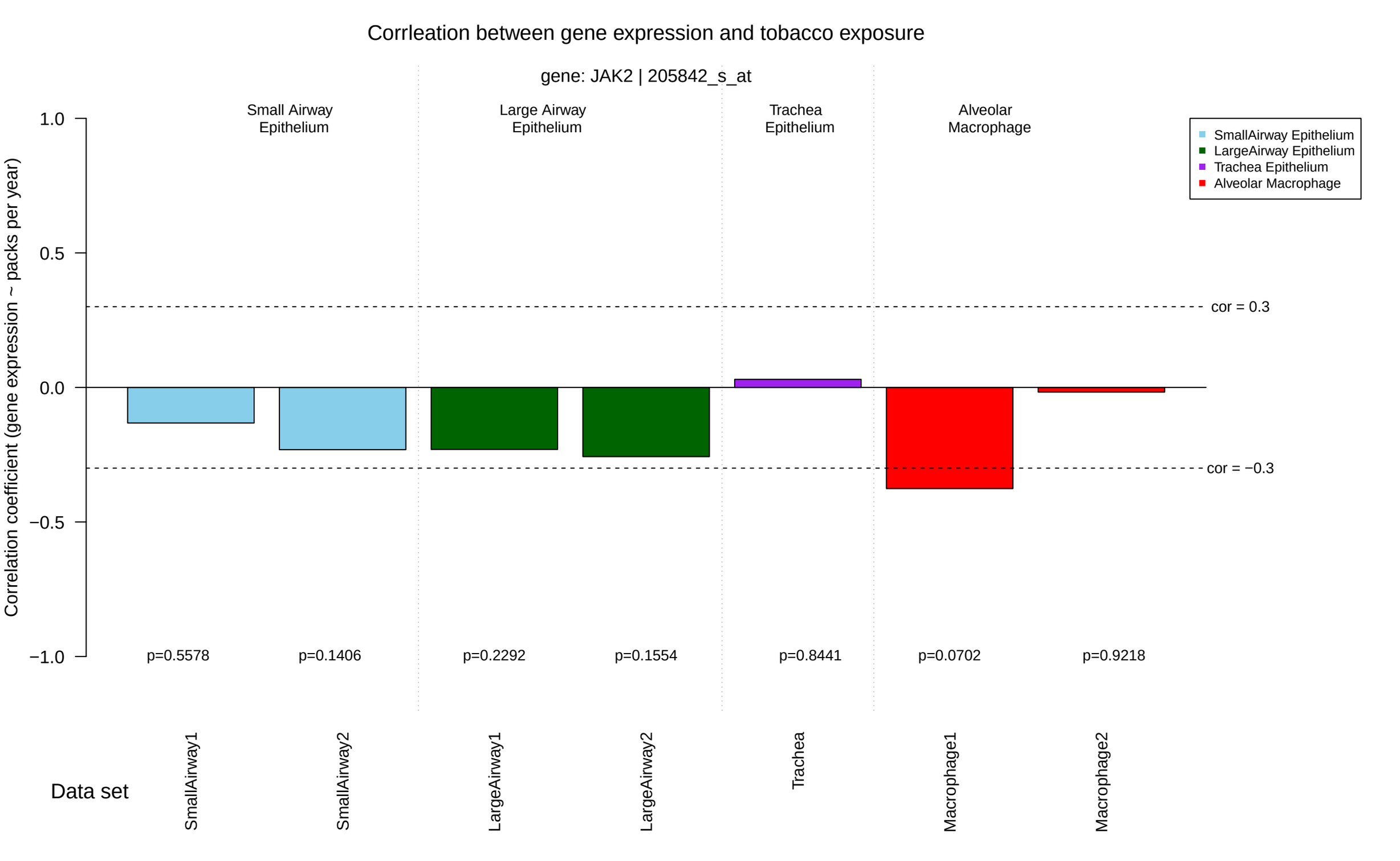

set: LargeAirway1 (p=0.0111)
JAK2 | 205842_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2126)
JAK2 | 205842_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4837)
JAK2 | 205842_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0048)
JAK2 | 205842_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.777)
JAK2 | 205842_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: JAK2 | 205842_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5578
p=0.1406
p=0.2292
p=0.1554
p=0.8441
p=0.0702
p=0.9218
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2